

Generation of Mean Flows and Jets on a Beta Plane and over Topography

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ABSTRACT

This paper proposes and discusses mechanisms whereby mean flows and jets are produced by differential rotation and by topographic effects. It is shown that, in general, a mean gradient of potential vorticity not only inhibits the cascade of energy to large scales but directly produces anisotropic structures. Scalings for this are examined on the β plane using ideas from classical phenomenology. The scalings are naturally anisotropic and predict the formation of zonal flows directly through a turbulent cascade. Numerical simulations and two-point closure calculations qualitatively confirm the predictions. Also, simulations of barotropic flow on the β plane can produce zonal jet structures of exceptional persistence over many eddy turnover times.

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Balanced mesoscale motion and stratified turbulence forced by convection

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SUMMARY

Numerical experiments are carried out with a large-eddy simulation model to investigate the production of stratified turbulence and gravity wave energy in the atmospheric mesoscale by deep convection. Relatively long integrations (typically about 2.5 days) are carried out, so that statistical equilibrium might be achieved for an atmosphere subject to fixed cooling and a ground surface maintained at constant temperature. The main simulations may be considered as representing convective activity in a cold airstream passing over warm seas. The transfer